REQUEST FOR PROPOSAL

Next-Generation BSS/OSS Platform Modernization

Vi Business Solutions | A Division of Vodafone Idea Limited

RFP Number	RFP-2026-VI-BSS-NXT-002
Issue Date	February 20, 2026
Submission Deadline	April 10, 2026, 5:00 PM IST
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1. Executive Summary

Vi Business Solutions (Vodafone Idea Limited) issues this RFP for comprehensive modernization of its BSS and OSS platforms, serving 210+ million subscribers across 22 telecom circles. The current landscape—built between 2012–2018—is fragmented, end-of-support, and unable to support 5G rollout, real-time monetization, and evolving customer experience standards. This initiative is board-approved under Vi's 'FutureReady' digital transformation programme.

The RFP seeks a single, integrated, cloud-native BSS/OSS stack covering:

- Real-time convergent charging and billing (prepaid, postpaid, enterprise)
- End-to-end order management and fulfillment automation
- Network inventory and resource management integrated with OSS
- AI/ML-driven customer lifecycle management and churn prediction
- Self-care portals and omnichannel customer engagement
- 5G network slice management and monetization capabilities
- Real-time revenue assurance and fraud management

2. Company Background

Legal Entity	Vodafone Idea Limited (Vi)
Headquarters	Mumbai, Maharashtra, India
Total Subscribers	~215 Million (Q3 FY2026)
Telecom Circles	22 circles across India
Network Infrastructure	470,000+ tower sites; 5G in 10 circles
Enterprise Customers	1.2 Million+ MSMEs and large enterprises
Annual Revenue	INR 43,000+ Crores (FY2025 estimate)
Employees	~18,000 direct employees

Current BSS/OSS Environment

Billing	Ericsson Charging System (ECS) + legacy convergent billing — on-premises
Order Management	Custom OMS with ~72-hour fulfillment cycles
CRM	Siebel v8.1, partially migrated to Salesforce for enterprise
Network Inventory	HP Service Activator + Excel-based shadow inventories
Self-Care	Three fragmented portals (consumer / SME / enterprise)
Revenue Assurance	SUBEX RAG v6.2 — no real-time capabilities

3. Project Overview & Objectives

This project entails a greenfield deployment of a cloud-native, microservices-based BSS/OSS platform to replace all legacy systems, managing the full subscriber lifecycle for 200M+ subscribers across consumer and enterprise segments. The platform must be production-ready by Q2 FY2027 via a phased migration.

Key Non-Functional Requirements

Scalability	215M subscribers; scalable to 300M without re-architecture
Availability (Charging)	99.999% (Five-Nines)
OCS Rating Latency	<50ms P99 at 1.5M TPS peak
Bill Run	10M accounts in <6 hours
Disaster Recovery	RPO <15 min; RTO <1 hour; Active-Active multi-region
Data Residency	100% within India; MeitY compliant cloud
Security	ISO 27001, PCI-DSS Level 1, TRAI, DPDP Act 2023
API Standards	TM Forum Open API (TMF) compliance mandatory

4. Scope of Work

Stream	Description
Stream A	Platform Design & Architecture — As-Is assessment, target architecture, cloud design, data migration strategy
Stream B	Software Deployment & Configuration — All BSS/OSS modules, 40+ integrations, performance & security testing
Stream C	Data Migration — 215M subscriber records, 24-month CDR history, inventory data, zero-revenue-leakage cutover
Stream D	Training & Change Management — 200 super-users, 3,500+ ops staff, runbooks and documentation
Stream E	Post-Go-Live Support — 90-day hypercare, 5-year AMS contract, L2/L3 SLA-governed support

Out-of-Scope

RAN equipment, core network (EPC/5GC) software, data center construction, MPLS upgrades, end-user device procurement, IT helpdesk.

5. Technical & Functional Requirements

BSS Module Requirements

Module	Key Specification
Online Charging System	3GPP Rel-17 Ro/Gy; <50ms P99; 1.5M TPS peak
Convergent Billing	Multi-currency/tax; 10M invoices in <6 hrs; GST e-invoice
CRM	360-degree view; AI-assisted workflows; <2s screen load
Order Management	Microservices OMS; <4-hour automation for SIM swap, plan change
Product Catalog	TM Forum SID-compliant; eSIM, 5G slices, FMC offers
Revenue Assurance	Real-time; <1-hour leakage detection

Module	Key Specification
Fraud Management	ML-based; <5-min alert-to-block for SIM cloning
Self-Care Portal	Web + iOS/Android; Hindi/English i18n; AI NLP chatbot

OSS Module Requirements

Module	Key Specification
Network Inventory	Dynamic discovery; 4G/5G-NSA/5G-SA; Ericsson, Nokia, Huawei, ZTE
Service Provisioning	NETCONF/YANG + REST; <60-sec SIM provisioning SLA
Fault Management	Real-time alarm correlation; <30-sec fault-to-ticket
Performance Management	15-min KPI intervals; Grafana/Kibana; ML anomaly detection
5G Slice Management	E2E slice lifecycle; GSMA NG.116 compliant; enterprise API exposure

Security & Compliance

Requirement	Standard
Data Encryption	AES-256 at rest; TLS 1.3 in transit; HSM key management
Authentication	MFA; SAML 2.0 / OAuth 2.0 / OIDC
Regulatory	TRAI, IT Act 2000, DPDP Act 2023, GSMA privacy frameworks
Audit Logging	Immutable logs; 36-month retention; SIEM (Splunk/QRadar)
Pen Testing	Bi-annual external; critical findings remediated within 30 days

6. Project Timeline & Milestones

RFP Process Schedule

RFP Release	February 20, 2026
Vendor Registration Deadline	March 5, 2026
Pre-Bid Conference	March 8, 2026 at 10:00 AM IST
Deadline for Written Questions	March 20, 2026
Proposal Submission Deadline	April 10, 2026 at 5:00 PM IST
Technical Evaluation	April 11–25, 2026
Vendor Demonstrations	April 28 – May 5, 2026
Contract Award	June 10, 2026
Programme Kickoff	July 1, 2026

Implementation Phases

Phase	Description & Target
Phase 0	Kickoff & Architecture Design — Aug 2026
Phase 1	Cloud Infra + Core OCS — Nov 2026
Phase 2A	CRM, OMS, Catalog (Enterprise) — Jan 2027
Phase 2B	Consumer Postpaid Billing (20M subs) — Mar 2027
Phase 3	Consumer Prepaid OCS (170M subs) — Jul 2027
Phase 4	OSS, RA/FM, Self-Care, AI/ML — Oct 2027
Phase 5	Legacy Decommission & Close — Dec 2027

7. Proposal Submission Instructions

Submit via Vi Vendor Portal: https://vendor.vibusiness.in/rfp-portal by April 10, 2026 at 5:00 PM IST. Physical submissions not accepted. Register by March 5, 2026.

Required Proposal Structure

Section	Content
Executive Summary	Max 5 pages
Company Profile & Qualifications	Full company background, certifications, and credentials
Technical Solution — BSS Modules	With compliance matrix
Technical Solution — OSS Modules	With compliance matrix
Integration & API Architecture	API design, TM Forum mappings, middleware
Cloud Infrastructure & Security Approach	Cloud design, DR, and security controls
Implementation Methodology	Gantt Chart & Team Structure
Data Migration Strategy	Approach for 215M subscriber records
Support & Managed Services Model	AMS model, SLAs, escalation procedures
Commercial Proposal / Pricing	Separate encrypted PDF
References & Case Studies	Min. 5 Tier-1 operator references + 3 case studies

Submission Format Requirements

Format	PDF only; A4, Arial/Calibri min 11pt, 25mm margins
File Naming	VendorName_Vi_BSS_RFP_2026_Technical.pdf
Max File Size	100 MB per file
Language	English only
Proposal Validity	180 days from submission deadline (until Oct 7, 2026)

8. Vendor Qualifications

Mandatory Pass/Fail Criteria

Non-compliance with any of the following criteria will result in immediate disqualification:

Requirement	Minimum Standard
Years in Business	Min. 10 years in BSS/OSS / telecom software
Telecom Revenue	Min. INR 500 Cr annual telecom revenue (last 3 FY)
India Presence	Dev centre + support ops in India; min. 500 FTE
Tier-1 Experience	Min. 2 live BSS/OSS implementations for 50M+ subscriber operators
Subscriber Scale	At least 1 reference with 100M+ subscribers migrated
OCS Throughput	Demonstrated 1M+ TPS in production
Data Migration	Successful migration of 50M+ subscriber records without billing disruption
5G Readiness	At least 1 live 5G slicing / SA core integration

Required Certifications

TM Forum Frameworx (Level 3+), GSMA FS.11/FS.19/FS.23, PCI-DSS Level 1, ISO 27001, ISO 9001, MeitY-empanelled cloud.

9. Evaluation Criteria

Category	Weight & Description
Technical Solution	40% — Architecture, TM Forum compliance, 5G readiness, security
Implementation Approach	20% — Project plan, migration strategy, risk management, team
Vendor Qualifications	15% — Tier-1 experience, certifications, India delivery capability
Support & Managed Services	10% — AMS model, SLA commitments, DR capabilities
Total Cost of Ownership (5-yr)	15% — Pricing competitiveness, commercial flexibility

Evaluation Process: Phase 1 — Qualification Screening (pass/fail) → Phase 2 — Technical & Commercial Scoring → Phase 3 — Demos & BAFO (top 3 vendors).

10. Pricing Requirements

Submit commercial proposal as a separate password-protected PDF. All prices in INR (excl. GST). Prices firm for 180 days from submission.

Cost Item	Details
Platform Software Licenses	BSS suite (OCS, Billing, CRM, OMS, Catalog) — per subscriber
OSS Software Licenses	Network Inventory, Provisioning, FM, PM, Slice Mgmt — per NE
Professional Services	Design, deployment, integration, testing — FTE-months
Data Migration Services	215M subscriber + CDR + inventory migration — lump sum
Training Services	Train-the-trainer + end-user — lump sum
Infrastructure Setup	Cloud onboarding, DevSecOps, CI/CD — lump sum
AMS (Annual)	Monthly fixed retainer + T&M; max 5% annual escalation
Cloud Infrastructure	Consumption-based IaaS estimate (Y1–Y5)

Volume tiers required for 200M / 250M / 300M subscribers. Provide milestone-linked payment schedule aligned to Phase gate completions.

11. Terms & Conditions

The selected vendor will sign a Master Services Agreement (MSA) with Vodafone Idea Limited. Key terms:

Provision	Details
Contract Duration	Fixed-price implementation (Phases 0–5, target Dec 2027) + 5-year AMS
Liability Cap	24 months AMS fees; uncapped for data breaches, IP infringement, gross negligence
Revenue Protection	Vendor liable for billing leakage >INR 1 Cr per incident attributable to platform defects
Data Protection	Full DPDP Act 2023 compliance; breach notification within 24 hours
IP Rights	Vi retains all configs, custom dev, data; vendor retains pre-existing IP
Escrow	Source code in escrow within 90 days of contract signing
Termination	Convenience: 120 days notice; Cause: 45-day cure period; Performance: 3 months SLA failure

SLA Service Credits (Charging/Billing Availability)

Availability	Service Credit
99.999%+	No credit (target)
99.99% – 99.998%	10% of monthly AMS fee
99.9% – 99.98%	25% of monthly AMS fee
Below 99.9%	50% of monthly AMS fee + right to invoke cure period

Insurance: Professional Indemnity INR 50 Cr, Cyber Liability INR 100 Cr, General Liability INR 25 Cr.
Disputes: Arbitration under Indian Arbitration & Conciliation Act 1996, seat Mumbai.

12. Appendices

Appendix	Contents
Appendix A	Required Vendor Forms (Cover letter, Vendor Info, NDA, Non-Collusion, Conflict of Interest, Reference Form, Compliance Matrix, Deviation Form)
Appendix B	Technical Annexures shared under NDA via Vendor Portal (Architecture Diagrams, NE Inventory, Data Volumes, Integration Catalogue, TM Forum Mapping, Regulatory Requirements, Performance Benchmarks)
Appendix C	Sample Service Level Agreement
Appendix D	Price Proposal Template (Excel)
Appendix E	Evaluation Scoring Sheet
Appendix F	Draft Master Services Agreement
Appendix G	Implementation Acceptance Criteria

Vi Business Solutions (Vodafone Idea Limited) thanks all prospective vendors for their interest.
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13. Detailed BSS Architecture & Module Specifications

This section expands on the BSS functional requirements with granular specifications for each module. Vendors must address each module individually and confirm compliance against every stated requirement using the provided response tables.

13.1 Online Charging System (OCS)

The OCS is the mission-critical real-time engine underpinning all prepaid and convergent charging for Vi's 215M+ subscribers. The replacement OCS must be cloud-native, horizontally scalable, and fully compliant with 3GPP Release 17 standards, supporting Diameter (Ro/Gy/Sy), REST, and gRPC interfaces. It must natively support 5G charging characteristics as defined in 3GPP TS 32.290 and TS 32.291.

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
OCS-001	The OCS shall support 3GPP Release 17 Ro interface (Diameter) for online charging of voice, SMS, and data services with full Credit-Control Request/Answer (CCR/CCA) message processing.		
OCS-002	The OCS shall support 3GPP Release 17 Gy interface for PCEF-based online charging and must interoperate with all PCEF/PCRF/PCF vendors deployed in Vi's network including Ericsson, Nokia, and Huawei.		
OCS-003	The OCS shall process a sustained throughput of 1,000,000 transactions per second (TPS) and support peak burst capacity of 1,500,000 TPS without performance degradation.		
OCS-004	OCS rating latency must not exceed 50ms at P99 under peak load conditions of 1.5M TPS, measured at the Diameter/HTTP2 interface boundary.		
OCS-005	The OCS shall support 5G CHF (Charging Function) interface as defined in 3GPP TS 32.291 (Nchf_ConvergedCharging service) for 5G SA core integration.		
OCS-006	The OCS shall support real-time balance management across multiple balance types (monetary, data, voice minutes, SMS, validity) with sub-second balance update propagation.		
OCS-007	The OCS shall support configurable re-rating of CDRs within a tolerance window of up to 30 days without manual intervention for tariff correction scenarios.		
OCS-008	The OCS shall implement threshold-based notifications (e.g., 80%, 90%, 100% usage) delivered via SMS, push notification, and in-app channels in under 5 seconds of threshold breach.		
OCS-009	The OCS shall support multi-dimensional rating based on time-of-day, day-of-week, location (home/roaming/visited network),		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
	content type, subscriber tier, and enterprise group hierarchies.		
OCS-010	The OCS must maintain active-active deployment across a minimum of 2 geographically separated data centres within India with automated failover completing in under 30 seconds.		

13.2 Convergent Billing System

The billing system must unify prepaid, postpaid, and hybrid billing across consumer and enterprise segments. It must support complex B2B bill structures, multi-currency invoicing for enterprise roaming, GST e-invoicing as mandated by the Government of India, and integration with Vi's ERP (SAP S/4 HANA) via standard APIs.

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
BLL-001	The billing system shall process a full billing run for 10,000,000 accounts within 6 hours, including mediation, rating, discounting, tax calculation, and invoice generation.		
BLL-002	The system shall support fully automated GST e-invoicing compliant with GSTN (Goods and Services Tax Network) IRN generation, e-way bills, and B2B invoice reconciliation as mandated by Indian tax law.		
BLL-003	The billing system shall support hierarchical enterprise billing with up to 10 levels of cost centre hierarchy, consolidated billing at parent level, and itemised billing at subsidiary level simultaneously.		
BLL-004	The system shall support multi-currency invoicing for enterprise customers with roaming charges, with automatic INR conversion using RBI-compliant exchange rates.		
BLL-005	The billing system shall support configurable bill cycles (monthly, bi-monthly, quarterly, custom anniversary date) with automatic proration for mid-cycle plan changes.		
BLL-006	The system shall provide a self-service bill dispute management module allowing CSRs and customers to raise, track, and resolve billing disputes with an SLA-tracked workflow.		
BLL-007	The billing system shall integrate with SAP S/4 HANA via certified APIs for revenue recognition, GL posting, accounts receivable, and collections management.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
BLL-008	The system shall generate digital bills in PDF, HTML5, and structured XML/JSON formats with multi-language support for English and Hindi.		
BLL-009	The billing system shall support real-time bill shock prevention by integrating with the OCS to provide mid-cycle spend alerts and configurable hard caps.		
BLL-010	The system shall support automated dunning workflows with configurable stages (reminder, warning, suspension, termination), channel preferences (SMS/email/IVR), and legal hold overrides.		

13.3 Customer Relationship Management (CRM)

The CRM platform must provide a unified 360-degree view of every subscriber — consumer or enterprise — across all touchpoints. It must support AI-driven Next Best Action (NBA) recommendations, seamless omnichannel engagement, and deep integration with the billing, OCS, and order management modules.

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
CRM-001	The CRM shall provide a unified 360-degree customer view aggregating subscription details, billing history (24 months), interaction history, device information, network usage, and active offers on a single screen.		
CRM-002	CRM screen load time for the 360-degree view must not exceed 2 seconds at P95 under concurrent usage by 3,500 active CSR agents.		
CRM-003	The CRM shall embed AI-driven Next Best Action (NBA) and Next Best Offer (NBO) recommendations at every customer interaction point, powered by a real-time ML inference engine.		
CRM-004	The CRM shall support omnichannel interaction tracking across in-store, call centre, web self-care, mobile app, WhatsApp, and social media channels with unified interaction timeline.		
CRM-005	The CRM shall support automated AI-based churn prediction scoring, updated daily at minimum, with configurable retention offer triggers for subscribers identified as high-risk.		
CRM-006	The system shall support full customer lifecycle management including prospect-to-subscriber onboarding, plan management, upgrade/downgrade, complaint management, and voluntary/involuntary churn processing.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
CRM-007	The CRM shall support enterprise account management with dedicated account hierarchies, multiple authorised contacts, credit limit management, and SLA tracking per enterprise contract.		
CRM-008	The CRM shall provide an integrated trouble ticket management module with automated routing, SLA tracking, escalation management, and real-time customer communication.		
CRM-009	The system shall integrate bidirectionally with the existing Salesforce Sales Cloud (enterprise segment) via certified REST APIs, ensuring no data duplication.		
CRM-010	The CRM shall provide configurable agent desktop with role-based access control, single sign-on (SSO), and full audit trail of all data accessed or modified by agents.		

13.4 Order Management System (OMS)

The OMS must orchestrate end-to-end service fulfilment from order capture through to network activation. It must support complex enterprise multi-service orders, automated fallout management, and real-time order status visibility for both agents and customers.

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
OMS-001	The OMS shall process standard consumer service orders (SIM swap, plan change, add-on activation) with end-to-end fulfilment in under 4 hours in fully automated happy-path scenarios.		
OMS-002	The OMS shall orchestrate complex enterprise multi-service orders (bulk SIM provisioning, VPN setup, IoT device onboarding) with configurable parallel and sequential task execution.		
OMS-003	The OMS shall expose a real-time order tracking API allowing customers and CSRs to query order status, estimated completion time, and current orchestration step.		
OMS-004	The OMS shall implement automated fallout detection and management, with configurable retry logic, fallout categorisation, and SLA-based escalation to manual queues.		
OMS-005	The system shall support a configurable product decomposition engine, translating commercial product catalogue items into network and IT system provisioning tasks automatically.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
OMS-006	The OMS shall integrate with all OSS provisioning systems (NETCONF/YANG, REST, SNMP) for automated network service activation without manual network engineer intervention for standard order types.		
OMS-007	The OMS shall support order jeopardy management with automated proactive notifications to customers and supervisors when orders are at risk of missing SLA commitments.		
OMS-008	The OMS shall maintain a complete and immutable order history with all state transitions, timestamps, and responsible system/agent identifiers for audit and regulatory compliance.		

14. Detailed OSS Architecture & Module Specifications

The OSS layer must provide complete visibility and control over Vi's 470,000+ tower sites across 22 circles, supporting 4G, 5G-NSA, and 5G-SA infrastructure from multiple vendors including Ericsson, Nokia, Huawei, and ZTE.

14.1 Network Inventory Management (NIM)

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
NIM-001	The NIM shall maintain a complete, accurate, and real-time inventory of all physical and logical network resources including RAN, core, transmission, and IT assets across all 22 circles.		
NIM-002	The NIM shall support automated dynamic discovery of network elements via SNMP v2c/v3, NETCONF, TL1, and vendor-specific APIs, with reconciliation against manually maintained records.		
NIM-003	The NIM shall support a multi-layer network model covering physical infrastructure (sites, cabinets, cards), logical topology (circuits, paths, VLANs), and service layer mapping.		
NIM-004	The NIM shall integrate with geographic information systems (GIS) to provide map-based visualisation of network assets, coverage areas, and fibre routes.		
NIM-005	The NIM shall support full lifecycle management of 5G NR equipment including gNB, AAU, RU/DU/CU split architecture, O-RAN compliant components, and virtual network functions (VNFs).		
NIM-006	Inventory data must be synchronised with OSS provisioning, fault management, and performance management modules in real-time, with change propagation latency under 60 seconds.		
NIM-007	The NIM shall provide a REST/TM Forum Open API interface allowing authorised third-party systems (ERP, workforce management, field service) to query and update inventory data.		

14.2 Service Provisioning & Activation

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
PRV-001	The provisioning system shall support SIM/eSIM activation on all Vi network elements within 60 seconds end-to-end from OMS trigger to network-confirmed active status.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
PRV-002	The system shall support NETCONF/YANG-based configuration management for all 5G NR network elements, with model-driven configuration templates for rapid service turn-up.		
PRV-003	The provisioning system shall support zero-touch provisioning (ZTP) for network elements, enabling automated initial configuration and software loading without on-site engineer intervention.		
PRV-004	The system shall maintain a configuration database (CDB) with full versioning and rollback capabilities, allowing restoration to any previous configuration state within 5 minutes.		
PRV-005	The provisioning system shall support bulk provisioning operations for enterprise customers (1,000+ SIMs simultaneously) with progress tracking and error reporting.		

14.3 Fault Management

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
FLT-001	The fault management system shall collect and correlate alarms from all network elements across all vendors (Ericsson, Nokia, Huawei, ZTE) in real-time via SNMP, NETCONF notifications, and vendor-specific interfaces.		
FLT-002	The system shall automatically generate a trouble ticket in the CRM/ticketing system within 30 seconds of a fault being confirmed, with pre-populated fault details, affected services, and impacted subscriber estimates.		
FLT-003	The fault management system shall include AI/ML-based root cause analysis (RCA) to correlate multiple related alarms into a single root cause event, reducing alarm storm volume by a target of 80% compared to raw alarm feed.		
FLT-004	The system shall support proactive network health monitoring with predictive fault detection using ML anomaly detection, generating alerts for network elements showing degradation patterns before outage occurs.		
FLT-005	The fault management system shall provide a real-time network health dashboard accessible to NOC engineers with drill-down capability from national view to individual cell level.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
FLT-006	The system shall integrate with Vi's existing SLA management framework to automatically flag fault events that breach or are projected to breach enterprise customer SLA commitments.		

14.4 Performance Management

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
PM-001	The performance management system shall collect KPI and KQI data from all network elements at 15-minute intervals, with configurable granularity down to 5-minute intervals for critical cells.		
PM-002	The system shall provide pre-built performance dashboards using Grafana/Kibana-compatible visualisation stacks, covering radio access, core, transmission, and service quality KPIs.		
PM-003	The performance management system shall support ML-based anomaly detection to identify statistically significant deviations from baseline performance patterns, with configurable sensitivity thresholds.		
PM-004	The system shall support long-term trend analysis and capacity planning reports with data retention of 24 months at 15-minute granularity and 36 months at daily aggregation.		
PM-005	Performance data shall be accessible via open REST APIs and support integration with Vi's existing data lake (AWS S3 or Azure ADLS) for advanced analytics and AI/ML workloads.		

14.5 5G Network Slice Management

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
SLC-001	The 5G slice management system shall support end-to-end lifecycle management of network slices including creation, modification, monitoring, and deletion per GSMA NG.116 specifications.		
SLC-002	The system shall support a minimum of 1,000 concurrent active network slice instances, each with independently defined QoS parameters, SLAs, and resource quotas.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
SLC-003	The slice management system shall expose an enterprise-facing API portal allowing enterprise customers to self-provision, modify, and monitor their dedicated network slices in near-real-time.		
SLC-004	The system shall support automated slice resource scaling based on real-time demand, integrating with the cloud orchestration layer to dynamically allocate/de-allocate compute and radio resources.		
SLC-005	The slice management system shall provide per-slice SLA monitoring and reporting, generating automated alerts and reports when slice KPIs deviate from agreed thresholds.		

15. Data Migration Strategy & Requirements

The data migration workstream is one of the highest-risk components of this programme, requiring the migration of over 215 million subscriber records, 24 months of CDR history, product catalogue configurations, network inventory data, and enterprise contract structures — all with zero revenue leakage and near-zero subscriber disruption.

15.1 Migration Scope

Subscriber Records	~215 Million (prepaid, postpaid, enterprise, IoT)
CDR History	24 months (estimated 15–20 TB compressed)
Active Balances	Real-time balance snapshot at cutover
Product Catalogue	~4,500 active product/offer configurations
Network Inventory	470,000+ network elements across 22 circles
Enterprise Contracts	~1.2 Million enterprise accounts with hierarchies
CRM Interaction History	12 months interaction history per customer

15.2 Data Migration Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
MIG-001	The vendor shall provide a detailed data migration strategy document covering source system extraction, data mapping, transformation rules, validation frameworks, cutover planning, and rollback procedures within 30 days of contract award.		
MIG-002	The migration approach shall support at minimum 3 full dress rehearsal migrations in a dedicated UAT environment before any production cutover activity commences.		
MIG-003	Migrated subscriber records must achieve 100% data integrity for all financially material fields (MSISDN, IMSI, account balance, active plan, bill cycle date, payment method) validated by automated reconciliation.		
MIG-004	The cutover strategy must maintain zero revenue leakage, defined as less than INR 1 Crore aggregate billing discrepancy attributable to migration activity per circle cutover event.		
MIG-005	CDR historical data migration must preserve all original fields including timestamp, called/calling party, duration, data volume, applied tariff, and revenue amount for 24 months post-cutover.		
MIG-006	The vendor must provide an automated data quality dashboard accessible to Vi's programme team showing migration progress, reconciliation scores, and		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
	exception counts in real-time during migration execution.		
MIG-007	The cutover plan must support a maximum maintenance window of 4 hours for prepaid cutover and 2 hours for postpaid cutover per circle, with full rollback capability executable within 1 hour.		
MIG-008	The vendor must provide dedicated on-site migration engineers in each of the 22 circles during circle-level cutover events, plus a central command and control function during all cutover windows.		
MIG-009	Post-migration, the vendor must operate parallel running validation for a minimum of 30 days, comparing legacy system outputs with new platform outputs for all billing events above INR 500.		
MIG-010	All migrated data must comply with DPDP Act 2023 requirements including data minimisation, purpose limitation, and consent status preservation from source systems.		

16. Security, Compliance & Data Governance Requirements

Vi Business Solutions operates under a stringent regulatory environment governed by TRAI, DoT, IT Act 2000, DPDP Act 2023, and international standards. All platform components must be designed with security-by-design principles and must maintain compliance with the following requirements without exception.

16.1 Data Security Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
SEC-001	All data at rest must be encrypted using AES-256 encryption with key management via a Hardware Security Module (HSM) that is FIPS 140-2 Level 3 certified.		
SEC-002	All data in transit must be protected using TLS 1.3 or higher. TLS 1.0, 1.1, and SSL 3.0 must be completely disabled across all platform components.		
SEC-003	Database access must enforce column-level encryption for all Personally Identifiable Information (PII) fields including MSISDN, Aadhaar-linked data, financial data, and location history.		
SEC-004	The platform must implement data masking and tokenisation for PII in all non-production environments (development, testing, UAT), with no real subscriber data permissible in lower environments.		
SEC-005	All encryption keys must be stored and managed in HSMs with dual control and split knowledge procedures. Key rotation must be automated and achievable without system downtime.		

16.2 Identity & Access Management

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
IAM-001	The platform must support Single Sign-On (SSO) via SAML 2.0 integration with Vi's corporate Identity Provider (Microsoft Azure AD), eliminating separate platform credentials for Vi staff.		
IAM-002	Multi-Factor Authentication (MFA) must be mandatory for all privileged access, administrative functions, and access to subscriber PII data, with TOTP and hardware token support.		
IAM-003	Role-Based Access Control (RBAC) must be implemented across all modules with a minimum of 50 predefined roles and support		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
	for custom role creation without vendor involvement.		
IAM-004	Privileged Access Management (PAM) controls must be implemented for all system administration and database access, with session recording for all privileged sessions.		
IAM-005	The platform must implement automatic session timeout of 15 minutes for inactive CSR sessions and 10 minutes for administrative sessions, with configurable override capability.		

16.3 Regulatory Compliance Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
REG-001	The platform must be fully compliant with TRAI Telecom Consumers Protection Regulations including Unsolicited Commercial Communication (UCC) blocking, DND registry integration, and consent management.		
REG-002	The platform must support full compliance with India's Digital Personal Data Protection Act 2023 (DPDP Act), including consent collection, storage, withdrawal management, and data principal rights fulfilment within statutory timelines.		
REG-003	The platform must maintain complete call data records (CDRs) in formats and for retention periods as specified by DoT/TRAI regulations (currently 1 year for CDRs, subject to regulatory change).		
REG-004	The platform must support Lawful Interception and Monitoring System (LIMS) integration as mandated under Indian Telegraph Act and DoT guidelines, with certified interfaces for authorised government agencies.		
REG-005	All cloud infrastructure hosting subscriber data must be located within the geographical boundary of India, compliant with MeitY data localisation requirements, with documented proof of residency available on demand.		
REG-006	The platform must be PCI-DSS Level 1 certified for all payment card processing functions, with annual QSA assessment and quarterly network scanning as part of ongoing compliance.		
REG-007	The platform must achieve and maintain ISO 27001:2022 certification within 12 months of go-live, with surveillance audits annually and recertification every 3 years.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
REG-008	Security incident and breach notification procedures must comply with CERT-In reporting requirements (within 6 hours of breach discovery) and DPDP Act requirements (72 hours to Data Protection Board).		

16.4 Audit & Logging Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
AUD-001	The platform must generate immutable audit logs for all data access, modification, creation, and deletion events involving subscriber PII, with tamper-evident storage using cryptographic hashing.		
AUD-002	Audit logs must be retained for a minimum of 36 months in a searchable format, with online access for the most recent 12 months and archived access for months 13–36.		
AUD-003	All audit logs must be forwarded in real-time to Vi's designated SIEM platform (Splunk or IBM QRadar) via standard syslog/CEF format for centralised security monitoring.		
AUD-004	The platform must provide pre-built compliance reporting dashboards for PCI-DSS, ISO 27001, and DPDP Act requirements, updated in near-real-time.		

17. Integration Architecture Requirements

The new BSS/OSS platform must integrate with over 40 external systems within Vi's IT and network landscape. All integrations must be API-first, standards-based, and designed for resilience with appropriate circuit breaker, retry, and dead-letter queue patterns.

17.1 Integration Standards & Governance

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
INT-001	All external-facing APIs must conform to TM Forum Open API (TMF) standards (minimum TMF versions 4.0+), with published API catalogue available on Vi's API developer portal.		
INT-002	The platform must support an API gateway layer providing authentication (OAuth 2.0), rate limiting, API versioning, SSL termination, and API analytics for all outbound and inbound integrations.		
INT-003	Integration patterns must support both synchronous REST/HTTP2 for real-time interactions and asynchronous event-driven patterns (Kafka/AMQP) for non-real-time data synchronisation.		
INT-004	The vendor must provide a comprehensive integration catalogue documenting all interfaces, API specifications (OpenAPI 3.0), message schemas, error codes, and SLA commitments per integration.		
INT-005	The platform must support an event streaming bus (Apache Kafka or equivalent) for high-volume event distribution to downstream analytics, AI/ML, and reporting systems.		

17.2 Key System Integrations

System	Integration Type	Key Data Flows
SAP S/4 HANA (ERP)	REST / IDoc via SAP Integration Suite	Revenue posting, accounts receivable, payment processing, GL entries
Salesforce (Enterprise CRM)	Bidirectional REST via Salesforce Connect	Enterprise account sync, opportunity data, contract status
GSTN e-Invoice Portal	REST API (NIC/GSTN portal)	IRN generation, e-way bill creation, invoice registration
Payment Gateway	PCI-DSS certified REST	Online payment processing, refunds, payment confirmation
PCRF/PCF (Policy)	Diameter Rx/Gx; HTTP/2 N5/N7	QoS policy installation, subscriber policy profiles
HLR/HSS/UDM	Diameter S6a/MAP; REST Nudm	Subscriber authentication data, roaming agreements

System	Integration Type	Key Data Flows
SMSC / Notification Platform	SMPP v3.4; REST; FCM/APNs	Billing alerts, OTP, marketing communications, push notifications
WhatsApp Business API	Meta Cloud API (REST)	Bill notifications, customer service interactions, chatbot
Splunk / QRadar (SIEM)	Syslog / CEF / REST	Security event logs, access logs, anomaly alerts

18. Cloud Infrastructure & DevSecOps Requirements

The platform must be deployed on MeitY-empanelled cloud infrastructure located entirely within India. Vi's preferred cloud provider(s) are AWS (Mumbai/Hyderabad regions) and/or Azure (Central India/South India regions). The vendor must support a cloud-agnostic Kubernetes-based deployment model to avoid vendor lock-in.

18.1 Cloud Infrastructure Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
INF-001	The platform must be deployed on MeitY-empanelled cloud infrastructure with all data, compute, and storage resources located within India's geographic boundaries without exception.		
INF-002	The platform must be designed for active-active multi-region deployment across a minimum of 2 Indian cloud regions to meet the 99.999% availability SLA for charging functions.		
INF-003	All platform components must be containerised using Docker and orchestrated via Kubernetes (EKS, AKS, or GKE), with infrastructure-as-code (Terraform/Pulumi) for all provisioning.		
INF-004	The platform must implement auto-scaling for all stateless components, capable of scaling from baseline to 150% of peak load within 10 minutes based on observable demand signals.		
INF-005	Disaster Recovery must be implemented with RPO < 15 minutes and RTO < 1 hour for all charging and billing functions, validated through quarterly DR drills with Vi participation.		
INF-006	The platform must implement a multi-tier network architecture with dedicated ingress/egress controls, WAF, DDoS protection (min. 10 Gbps protection), and network micro-segmentation.		

18.2 DevSecOps & CI/CD Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
DEV-001	The vendor must establish a fully automated CI/CD pipeline (Jenkins/GitLab CI/GitHub Actions) for all platform components, enabling tested code changes to reach production within 4 hours of approval.		
DEV-002	All code must undergo automated static application security testing (SAST) and software composition analysis (SCA) as part		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
	of every pipeline execution, with no Critical-severity findings permissible in production branches.		
DEV-003	Container images must be scanned for vulnerabilities using an automated scanner (Trivy/Twistlock/Prisma Cloud) with critical vulnerabilities triggering pipeline failure and automatic build rejection.		
DEV-004	The platform must implement GitOps practices for all configuration and infrastructure changes, with all changes requiring peer review and approval before execution against production environments.		
DEV-005	The vendor must provide Vi's technical team full access to CI/CD pipelines, Git repositories (for Vi-owned customisations), and deployment dashboards from Day 1 of programme kickoff.		

19. Programme Governance & Delivery Framework

Given the scale and strategic importance of this programme, a robust governance structure is mandatory. The vendor must establish and resource the following governance framework from programme kickoff.

19.1 Governance Structure

Governance Body	Cadence	Purpose & Attendees
Programme Steering Committee	Monthly	Strategic direction, phase gate approvals, budget decisions. Vi CTO + CDO + CFO; Vendor CEO/CTO
Programme Management Office (PMO)	Weekly	Programme status, risk/issue review, milestone tracking. Vi Programme Director + Vendor Programme Director
Technical Architecture Board (TAB)	Bi-weekly	Architecture decisions, integration design review, security sign-off. Vi Enterprise Architects + Vendor Solution Architects
Stream-Level Working Groups	Weekly	Stream-specific delivery tracking, dependency management. Vi Stream Owners + Vendor Stream Leads
Security & Compliance Board	Monthly	Security posture review, compliance status, vulnerability management. Vi CISO + Vendor Security Lead

19.2 Risk Management Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
RISK-001	The vendor must maintain a live programme risk register with a minimum of a 3x5 probability/impact matrix, updated weekly and shared with Vi PMO.		
RISK-002	The vendor must prepare and maintain a formal Business Continuity Plan (BCP) and Crisis Management Plan specifically for the migration programme, approved by Vi within 60 days of kickoff.		
RISK-003	The vendor must identify and assign a dedicated risk owner for each P1/P2 risk item, with documented mitigation plans, contingency plans, and trigger conditions.		
RISK-004	The vendor must conduct a formal Dependency Management Review at each phase gate, mapping all internal and external dependencies with critical path analysis.		

19.3 Acceptance Testing Framework

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
UAT-001	The vendor must deliver a comprehensive Test Strategy document covering System Integration Testing (SIT), User Acceptance Testing (UAT), Performance & Load Testing, Security Testing, and Disaster Recovery Testing within 30 days of architecture sign-off.		
UAT-002	Performance testing must simulate peak load conditions equivalent to 1.5x the projected maximum daily subscriber activity, conducted in an environment that mirrors production infrastructure at minimum 50% scale.		
UAT-003	Each phase gate must include a formal Acceptance Test Pack (ATP) reviewed and signed off by Vi's Test Manager before phase completion can be declared.		
UAT-004	The vendor must conduct a dedicated Billing Accuracy UAT covering a minimum of 500 complex test scenarios per billing module, including multi-service bundles, proration, discounts, and tax calculations.		

20. Support & Managed Services Requirements

Post go-live, the selected vendor will be required to provide a comprehensive Application Management Services (AMS) engagement for a minimum of 5 years, covering all platform modules delivered under this RFP.

20.1 Service Levels

Priority	Definition	Initial Response SLA	Resolution SLA
P1 - Critical	Charging/billing platform down; revenue-impacting outage	15 minutes	4 hours with 30-min progress updates
P2 - High	Significant functionality impaired; >10% subscriber impact	30 minutes	8 hours with hourly progress updates
P3 - Medium	Functionality degraded; workaround available	2 hours	3 business days
P4 - Low	Minor issue; cosmetic or informational	4 business hours	10 business days

20.2 AMS Requirements

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
AMS-001	The vendor must provide a dedicated 24x7x365 L2/L3 support team with minimum 20 FTEs dedicated to Vi's AMS engagement, co-located in India.		
AMS-002	The AMS team must include dedicated specialists for OCS, Billing, CRM, OSS, Security, and Database Administration available within 30 minutes of P1 alert.		
AMS-003	The vendor must provide a dedicated Customer Success Manager (CSM) as a single point of accountability for Vi, conducting monthly service reviews and quarterly business reviews.		
AMS-004	Proactive monitoring must cover 100% of platform components with automated alerting configured to detect anomalies and notify the AMS team before subscriber or revenue impact occurs.		
AMS-005	The vendor must provide a comprehensive monthly service report covering availability, incident statistics, problem management status, change management summary, and capacity utilisation.		
AMS-006	Software patches for critical security vulnerabilities (CVSS >= 9.0) must be deployed within 72 hours of vendor release.		

Req ID	Requirement Description	Compliance (Yes / No / Partial)	Vendor Remarks / Solution Approach
	High severity patches (CVSS 7.0–8.9) within 14 days.		
AMS-007	The AMS contract must include a minimum of 2 major platform version upgrades per year, executed during agreed maintenance windows with zero subscriber-impacting downtime.		

21. Vendor Profile & Commercial Summary — Fill in the Blanks

This section must be completed by the responding vendor in full. All fields are mandatory. Vi will use this section for rapid comparative evaluation across shortlisted vendors. All fields must reflect the proposed solution for this specific engagement (Vi BSS/OSS Modernisation).

Note: Yellow-shaded cells throughout this RFP and all fill-in fields below must be completed by the vendor. Incomplete submissions will be scored as zero for the respective evaluation criterion.

21.1 Vendor Identity & Corporate Standing

Field	Vendor Response (Fill in)
Legal Registered Name of Proposing Entity	
Registered Office Address (India)	
Name and Title of Authorised Signatory	
Year of Incorporation	
GST Registration Number	
Number of FTEs in India (Total)	
Number of FTEs dedicated to Telecom BSS/OSS (India)	
Annual Telecom Revenue (FY2025, INR Crores)	
Total Company Revenue (FY2025, INR/USD)	
Credit Rating (if applicable)	

21.2 Platform Performance Commitments

The vendor must complete the table below with the guaranteed performance metrics for the proposed platform as it will be deployed for Vi's environment.

Performance Parameter	Vendor Committed Value
The proposed BSS/OSS platform will support a maximum of [___] concurrent transactions per second (TPS) for the OCS.	
The OCS rating latency guarantee at P99 under peak load of 1.5M TPS will be [___] milliseconds.	
The maximum number of subscribers the platform can support without architectural re-design is [___] Million.	
A full billing run for 10 million accounts will complete within [___] hours.	
Guaranteed platform availability (SLA) for charging functions is [___]% per month.	
Disaster Recovery RTO for full platform restoration is [___] minutes.	

Performance Parameter	Vendor Committed Value
Disaster Recovery RPO (maximum data loss window) is [___] minutes.	
SIM provisioning end-to-end activation SLA from OMS trigger is [___] seconds.	
Fault-to-ticket automated generation time from alarm confirmation is [___] seconds.	
CRM 360-degree view screen load time at P95 under 3,500 concurrent agents is [___] seconds.	

21.3 Implementation Timeline Commitments

Timeline Parameter	Vendor Committed Value
The total estimated time to complete Phase 1 (Cloud Infrastructure + Core OCS) from programme kickoff is [___] months.	
The estimated number of months from contract award to first production subscriber (Phase 2B Go-Live) is [___] months.	
The total programme duration from kickoff to legacy decommission (Phase 5 completion) will be [___] months.	
The number of dedicated FTEs the vendor will deploy on this programme at peak staffing is [___] FTEs.	
The number of dress rehearsal migrations to be conducted in UAT before any production cutover is [___] rehearsals.	
The maximum maintenance window required for each circle-level subscriber migration cutover is [___] hours.	
Time from contract signing to delivery of the detailed Architecture Design Document (ADD) is [___] weeks.	

21.4 Commercial Summary

Provide indicative commercial figures in the table below. Detailed breakdown must be provided in the separate encrypted commercial PDF per Section 10.

Commercial Parameter	Indicative Value (INR, excl. GST)
Total BSS Software Licensing Cost (for 200M subscribers, Year 1)	
Total OSS Software Licensing Cost (per NE, Year 1)	
Total Professional Services — Implementation (Phases 0–5)	
Total Data Migration Services (lump sum)	
Training Services (total lump sum)	
Cloud Infrastructure — Year 1 IaaS Estimate	
Annual Managed Services (AMS) Fee — Year 1	

Commercial Parameter	Indicative Value (INR, excl. GST)
Total 5-Year Total Cost of Ownership (TCO)	
Payment Milestone — Amount due at Contract Signing (%)	
Payment Milestone — Amount due at Phase 1 Go-Live (%)	

21.5 Reference Engagements

Provide the following details for a minimum of 3 reference engagements. Vi reserves the right to contact all references provided.

Reference Field	Reference 1 | Reference 2 | Reference 3
Operator Name & Country	
Number of Subscribers on Platform	
Modules Deployed	
Go-Live Date	
Reference Contact Name & Email	
Is this a 100M+ subscriber reference? (Y/N)	

21.6 Certifications & Standards Compliance

Certification / Standard	Currently Held? (Y/N)	Certificate Reference / Expiry
TM Forum Frameworx Level 3+		
ISO 27001:2022		
ISO 9001:2015		
PCI-DSS Level 1		
GSMA FS.11 / FS.19 / FS.23		
MeitY-Empanelled Cloud (if self-hosted)		
3GPP Release 17 OCS Compliance (certified or self-attested)		
GSMA NG.116 (5G Slicing)		

END OF REQUEST FOR PROPOSAL

Vi Business Solutions (Vodafone Idea Limited) thanks all prospective vendors for their interest in this engagement.

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